IoT Device Fingerprinting and Rogue Device Detection— A Machine Learning-Based Framework for Enterprise Network Security

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Abstract—This paper addresses the challenge of IoT traffic behavior fingerprinting and rogue traffic detection in enterprise networks by proposing a hybrid machine learning framework that combines supervised and unsupervised techniques. The framework integrates traffic behavior fingerprinting, anomaly detection, and decision fusion mechanisms within a Zero Trust architecture. It was evaluated using the CICIoT2023 dataset, achieving improved detection performance compared to standalone models. The results demonstrate that the hybrid approach enhances accuracy and detection of previously unseen traffic patterns. This study contributes to a scalable and practical solution for enterprise network security.

Keywords—IoT security, traffic fingerprinting, anomaly detection, machine learning, open-set detection, Zero Trust

1. INTRODUCTION

1.1 Background & Context

The rapid expansion of Internet of Things (IoT) devices in enterprise environments has significantly increased the attack surface and complexity of network security [1]. IoT ecosystems generate large volumes of heterogeneous network traffic, often involving resource-constrained devices operating with limited security controls. As a result, traditional security mechanisms based on static identifiers, such as MAC or IP addresses, are increasingly ineffective in identifying and securing IoT devices [2].

In parallel, the widespread adoption of encrypted communication protocols has reduced the effectiveness of deep packet inspection techniques, limiting visibility into payload-level data. Consequently, there is a growing need for approaches that rely on traffic behavior rather than static identifiers or payload inspection to achieve reliable detection and classification in modern enterprise networks.

1.2 Problem Statement

Accurate identification of IoT devices and detection of rogue or unauthorized entities remain critical challenges in enterprise network security. Existing approaches face several limitations in detecting unknown or evolving threats [3]:

1.2.1 Device identification limitations: MAC and IP addresses can be spoofed, making them unreliable for authentication and tracking.

1.2.2 Encrypted traffic visibility gap: Increasing use of encryption restricts payload inspection, reducing detection effectiveness.

1.2.3 Detection–response disconnect: Many solutions focus solely on detection without integrating decision-making and enforcement mechanisms. In practice, due to dataset constraints such as those present in CICIoT2023, traffic behavior must be used as a proxy for device identity (traffic-level representation. While this introduces abstraction, it enables robust classification and anomaly detection in encrypted environments where payload inspection is not feasible.

1.3 Research Gap

Despite advances in machine learning–based intrusion detection, current solutions remain fragmented and limited in scope. The key gaps identified in the literature include:

- Lack of integrated frameworks combining classification, anomaly detection, and response mechanisms [4].
- Limited support for open-set conditions, where unknown or previously unseen traffic must be detected.
- Absence of decision fusion strategies that combine multiple model outputs into a unified risk-based decision.
- Weak alignment with Zero Trust security principles, particularly in enforcement and continuous monitoring.

These gaps highlight the need for a unified framework capable of addressing both known and unknown threats while integrating detection with actionable security responses.

1.4 Contributions

This study proposes a comprehensive machine learning–based framework to address the identified gaps. The main contributions are:

- A hybrid supervised–unsupervised framework for IoT traffic behavior fingerprinting.
- An open-set rogue traffic detection mechanism for identifying previously unseen patterns.
- A decision fusion engine that integrates classification and anomaly detection outputs.

- Alignment with Zero Trust architecture principles, including segmentation and enforcement [5].
- Experimental validation using the CICIoT2023 dataset, demonstrating improved performance over standalone models [6].

2. LITERATURE REVIEW

This section provides a structured overview of existing research on IoT traffic fingerprinting, machine learning-based classification, anomaly detection, and open-set recognition. The review establishes the theoretical and methodological foundations for the proposed hybrid framework and identifies key limitations in current approaches.

2.1 IoT Traffic Fingerprinting

IoT Traffic Fingerprinting techniques aim to identify devices based on network behavior rather than static identifiers [2], [7]. Prior work has demonstrated that traffic patterns, timing characteristics, and protocol usage can uniquely represent device behavior. However, most approaches assume access to device-level labels, which are not always available in real-world datasets.

2.2 ML-Based Traffic Classification

Machine learning models such as Random Forest, XGBoost, and deep learning architectures have been widely used for traffic classification [2], [3]. These approaches achieve high accuracy in closed-set scenarios where all classes are known during training. However, their performance degrades when encountering previously unseen traffic classes.

2.3 Anomaly Detection Techniques

Unsupervised methods, including clustering algorithms (e.g., K-Means, DBSCAN) and autoencoder-based models, have been applied to detect anomalous network behavior [3]. Frameworks such as Kitsune demonstrate the effectiveness of autoencoders in identifying deviations from normal traffic patterns [8]. While these methods are effective for anomaly detection, they lack classification capabilities and may produce high false positive rates.

2.4 Open-Set Recognition in Cybersecurity

Open-set recognition addresses scenarios where unknown classes may appear during testing. In cybersecurity contexts, this is particularly relevant for detecting novel attacks. Existing research highlights the importance of combining classification and anomaly detection to improve robustness under open-set conditions [4].

2.5 Critical Gap Analysis

Existing studies in IoT traffic analysis predominantly focus on either supervised classification or unsupervised anomaly detection in isolation [4], [5]. While these approaches have demonstrated strong performance within their respective domains, they are limited in their ability to operate effectively under dynamic, real-world conditions where unknown traffic patterns and emerging threats are common. Most frameworks assume closed-set environments and lack mechanisms to detect previously unseen traffic classes.

Furthermore, there is a notable absence of unified architectures that integrate classification, anomaly detection, and decision-making into a cohesive framework. Existing solutions also provide limited support for operational security integration, particularly in the context of Zero Trust principles, where continuous monitoring, risk-based decision-making, and enforcement are critical.

To address these limitations, this study proposes a hybrid framework that combines supervised traffic classification with unsupervised anomaly detection under open-set conditions. The framework further incorporates a decision fusion mechanism to generate unified risk-based outputs and integrates alerting and segmentation capabilities aligned with Zero Trust architecture, enabling a practical and scalable approach for enterprise IoT security.

3. PROBLEM STATEMENT / RESEARCH MOTIVATION

The increasing reliance on IoT devices in enterprise networks introduces significant security challenges, particularly in identifying devices and detecting unauthorized or malicious activity. Traditional approaches are insufficient due to spoofing, encryption, and the dynamic nature of IoT environments.

The motivation of this research is to develop a scalable and practical framework that can:

- Reliably classify known traffic patterns.
- Detects unknown or anomalous traffic under open-set conditions.
- Integrate detection outputs into actionable security decisions.
- Support enforcement mechanisms aligned with Zero Trust principles.

This research is both academically relevant, addressing gaps in hybrid detection frameworks, and practically significant for improving enterprise network security operations.

4. OBJECTIVES / RESEARCH QUESTIONS

4.1 Research Objectives

The primary objective of this research is to propose a hybrid machine learning framework for IoT traffic behavior fingerprinting and rogue traffic detection under open-set conditions. Specific objectives include:

4.1.1 Objective 1— Supervised Classification (Layer 3): Design and implementation of a supervised multi-class classification model for accurate identification of known IoT traffic behaviors.

- Implemented in Section 6
  - 6.3–6.5: Data prep + splits
  - 6.7: Random Forest, XGBoost, CNN

- Evaluated in Section 7
  - 7.1 Accuracy
  - 7.1 Precision / Recall / F1
  - 7.1 ROC-AUC
  - 7.1 Confusion Matrix

4.1.2 Objective 2— Unsupervised Detection (Layer 4): Integration of unsupervised anomaly detection techniques to identify unknown or previously unseen traffic patterns.

- Implemented in Section 6
  - 6.7: K-Means, DBSCAN, Autoencoder
  - 6.3.4: Unknown holdout
- Evaluated in Section 7
  - 7.2 Clustering behavior
  - 7.2 Open-set detection
  - 7.3 TPR / FPR
  - 7.3 ROC

4.1.3 Objective 3— Decision Fusion (Layer 5): Design of a decision fusion mechanism that combines supervised classification outputs with anomaly detection scores to improve detection accuracy and reduce false positives.

- Implemented in Section 6.8 Fusion logic description
- Evaluated in Section 7
  - 7.3 Detection of unseen traffic
  - 7.3 TPR vs FPR

4.1.4 Objective 4— Evaluation under Open-Set Conditions (Core Contribution): Evaluation of the proposed framework using classification and anomaly detection metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC, TPR, and FPR) under open-set conditions.

- Implemented in Section 6
  - 6.5.4: Unknown holdout design
  - 6.8: Metrics definition
- Evaluated in Section 7
  - 7.3 Detection of unseen traffic
  - 7.3 TPR vs FPR

4.1.5 Objective 5— Zero Trust Alignment (Layers 6-8): Demonstration of the framework's alignment with Zero Trust principles, incl. continuous monitoring, risk-based decision-making, and enforcement mechanisms.

- Implemented in Section 5
  - SIEM (Layer 6)
  - Segmentation (Layer 7)
  - Continuous learning (Layer 8)
- Evaluated in Section 8
  - Integration
  - Real-world applicability
  - Zero Trust alignment

Each objective is directly mapped to the corresponding methodology components (Section 6) and evaluated through experimental results (Section 7), ensuring full traceability between design, implementation, and validation.

4.2 Research Questions

This study is guided by the following research questions:

4.2.1 RQ1 (Performance Comparison): To what extent does a hybrid supervised–unsupervised framework improve detection performance compared to standalone models?

4.2.2 RQ2 (Open-Set Generalization): How effectively can the framework detect previously unseen traffic patterns under open-set conditions? puts affect detection accuracy and false positive rates?

4.2.3 RQ4 (Practical Applicability): How well does the proposed framework align with enterprise security requirements under a Zero Trust architecture?

5. PROPOSED FRAMEWORK / ARCHITECTURE

5.1 System Architecture Overview

As shown in Fig. 1, the proposed eight-layer architecture provides an end-to-end framework for IoT traffic behavior fingerprinting and rogue traffic detection. The framework integrates supervised classification of known traffic patterns with unsupervised anomaly detection for previously unseen traffic under open-set conditions.

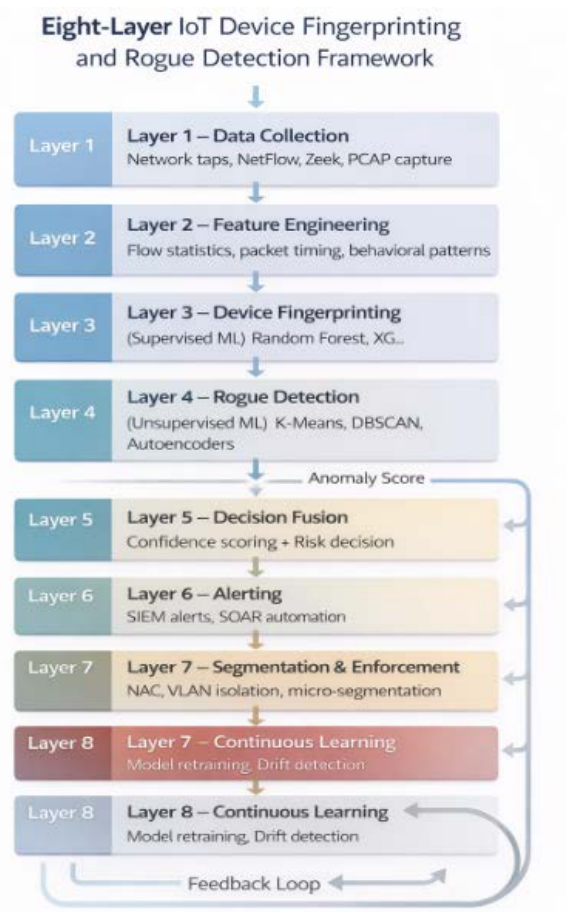

Fig. 1. Eight-layer IoT device fingerprinting and rogue device detection framework.

A decision fusion layer combines model outputs to generate unified risk-based decisions, which are

subsequently operationalized through alerting and enforcement mechanisms within a Zero Trust security model. Furthermore, a continuous learning layer enables feedback-driven model adaptation, improving detection performance over time.

The architecture supports both closed-set classification of known traffic and open-set detection of unknown traffic, ensuring robustness in dynamic enterprise environments. This layered design enables modular evaluation, scalability, and flexibility, allowing adaptive weighting or selective reliance on individual detection layers based on empirical performance.

5.2 Layered Design Explanation

The functional roles of each layer within the proposed architecture are summarized in Table 1.

Layer	Function
L1–2	Data Collection & Feature Engineering
L3	Supervised Fingerprinting
L4	Unsupervised Rogue Detection
L5	Decision Fusion Engine
L6	Alerting (SIEM logic)
L7	Network Segmentation / Enforcement
L8	Continuous Learning

Table 1: Layered Design

6. METHODOLOGY

6.1 Research Design

This study followed an end-to-end machine learning–driven research design to develop and evaluate an integrated framework for IoT traffic behavior fingerprinting and rogue traffic detection. The methodology combined supervised and unsupervised learning techniques within a structured eight-layer architecture aligned with Zero Trust principles [5].

The approach consisted of: (i) dataset preparation and feature engineering, (ii) supervised classification of known IoT devices, (iii) unsupervised anomaly detection for previously unseen traffic patterns, and (iv) decision fusion to generate unified risk-based outputs. This design ensured both high classification accuracy for known devices and robust detection of previously unseen devices in open-set conditions.

6.2 Dataset Description

The framework was evaluated using the CICIoT2023 dataset [6], a large-scale IoT network traffic dataset developed by the Canadian Institute for Cybersecurity at the University of New Brunswick. The full dataset contains over 47 million labeled network flows captured from 105 heterogeneous IoT devices, with 33 distinct attack types across seven attack categories (DDoS, DoS, Reconnaissance, Web-based, Brute Force, Spoofing, and Mirai variants), in addition to benign traffic.

For this study, a representative subsample of 4,016,753 flows was drawn from six merged CSV files (Merged01, 02, 05, 09, 32, and 63). These files are pre-shuffled and contain all attack categories in proportion to their occurrence in the full dataset, ensuring that the subsample preserves the original class distribution. Each flow is described by 39 numerical features and a single categorical label, for a total of 40 columns.

The 39 features capture flow-level statistical and behavioral characteristics, including packet header metadata (header length, time-to-live, protocol type), TCP flag counts (FIN, SYN, RST, PSH, ACK, ECE, CWR), protocol indicators (HTTP, HTTPS, DNS, Telnet, SMTP, SSH, IRC, TCP, UDP, DHCP, ARP, ICMP, IGMP, IPv, LLC), and aggregate statistical descriptors (total bytes, minimum, maximum, average, standard deviation, total size, inter-arrival time, packet count, and variance). These features are derived entirely from flow metadata and do not require deep packet inspection, which makes the framework applicable to encrypted traffic environments.

A total of 114 rows containing NaN or infinite values were removed during preprocessing, representing a negligible proportion of the dataset. Following preprocessing, the dataset was further refined through class balancing and partitioning procedures, as described in Section 6.3 and Section 6.5.

6.2.1 Reframing from Device-Level to Traffic-Class Fingerprinting

It is important to note that CICIoT2023 does not include a per-device identifier in its merged CSV files. The label column identifies the traffic class of each flow (benign or one of 33 attack types) rather than the source device. Consequently, the framework presented in this paper performs traffic behavior fingerprinting and rogue traffic class detection, using attack-class behavior as a proxy for the broader concept of device behavior fingerprinting. This is consistent with the established practice in the CICIoT2023 literature [6], where the dataset's strength lies in distinguishing benign and malicious traffic patterns rather than identifying individual devices. The hybrid supervised plus unsupervised approach proposed here remains directly transferable to device-level fingerprinting in datasets where per-device labels are available, and we identify this as a future work direction. This reframing does not affect the validity of the proposed framework, as traffic behavior remains a reliable proxy for device-level activity in IoT security research.

6.2.2 Class Distribution and Selection

The 4.02 million-flow subsample contains 34 distinct classes: one benign class and 33 attack classes. The class distribution is heavily imbalanced, with the ten most frequent classes (all DDoS and DoS variants) accounting for approximately 85 percent of the total flows. Benign traffic originally comprised approximately 94K flows and was subsequently capped at 30K within the known-class training subset to ensure balanced representation

across classes. A long tail of rare attack classes contains fewer than 250 flows each. This natural imbalance and the presence of rare classes make CICIoT2023 well-suited for evaluating open-set detection performance, since the rare classes provide a credible source of unknown traffic samples for Layer 4 evaluation.

6.3 Data Preprocessing

Data preprocessing was conducted in five stages to ensure model consistency, balanced class representation, and methodologically sound open-set evaluation. All reported dataset statistics were derived from the initial data inventory and preprocessing stages, with final partition sizes confirmed following class balancing and dataset splitting.

6.3.1 Stage 1— Data Loading and Inventory

The six merged CSV files were loaded using pandas in Google Colab Pro and concatenated into a single dataframe of 4,016,753 raw rows and 40 columns. Column types were verified: 39 numerical features (30 float and 9 integer) and one categorical label column. The combined dataset occupied approximately 1.4--1.6 GB in memory.

6.3.2 Stage 2— Missing Value Handling

An inventory of missing values identified 60 NaN entries across the combined dataset; all confined to two statistical columns (standard deviation and variance). These NaN values arise when a flow contains only a single packet, rendering variance and standard deviation undefined. The affected rows were removed, representing less than 0.004 percent of the dataset and having negligible impact on the overall distribution. In total, 114 rows containing NaN or infinite values were removed during preprocessing.

Additionally, infinite values resulting from zero-duration flows were replaced with NaN and subsequently removed, consistent with known CICIoT2023 data characteristics.

6.3.3 Stage 3— Class Selection and Per-Class Capping

To address the severe class imbalance and prevent the supervised models from memorizing the most frequent DDoS variants, a per-class cap of 30,000 flows was applied to all known classes. Classes with fewer than 30,000 flows were retained in their entirety.

This capping reduced the known-class portion of the dataset from 4,016,753 flows to approximately 611,566 flows after class capping and dataset partitioning while preserving representation across all attack categories. Benign flows were retained in full (approximately 94K flows prior to capping) to maximize the training data available to the Layer 4 Autoencoder, which trains exclusively on benign traffic.

6.3.4 Stage 4— Open-Set Holdout Selection

Five rare attack classes were selected as the unknown holdout set, removed entirely from the training and validation data, and reserved exclusively for Layer 4 open-set evaluation. The selected classes are:

- BACKDOOR_MALWARE (131 flows) — distinct malware-class behavior.
- BROWSERHIJACKING (248 flows) — web-layer attack with distinct traffic profile.
- XSS (177 flows)— web injection attack.
- RECON-PINGSWEEP (85 flows) — reconnaissance behavior distinct from main DDoS classes.
- UPLOADING_ATTACK (48 flows) — file upload exploitation.

These five classes were selected for two reasons. First, each is rare enough that its removal does not materially affect the supervised training distribution. Second, together they span four distinct attack categories (malware, web injection, web hijacking, reconnaissance, and file upload), providing the diversity needed to evaluate open-set generalization rather than detection of a single anomalous pattern.

The combined holdout contains 1,455 flows across 5 classes, which is sufficient to compute meaningful true positive and false positive rates while remaining a minority of the overall test set.

6.3.5 Stage 5— Feature Normalization

All 39 numerical features were standardized using z-score normalization (zero mean, unit variance), with normalization parameters fit on the training set only and applied to the validation, test, and unknown holdout sets. This prevents information leakage from the test data into model training and ensures consistent feature scaling across all four data partitions. The fitted normalization parameters persisted and reused during inference to ensure reproducibility across all experimental stages.

6.4 Experimental Setup

The experimental environment was implemented using Google Colab (Python-based environment) with standard machine learning libraries. The setup included:

- Programming language: Python.
- Libraries: Scikit-learn, Pandas, NumPy, TensorFlow/Keras (for deep learning models).
- Execution environment: Google Colab (cloud-based computing).
- Hardware: CPU/GPU-enabled environment (as available in Colab).

This setup enabled scalable model training and evaluation on large IoT datasets. Model development and evaluation were conducted using a unified notebook pipeline (03_modeling.ipynb), ensuring consistent and reproducible implementation across Layers 3–5.

The use of a shared notebook environment ensured alignment across team members and facilitated reproducible experimentation workflows.

6.5 Data Splitting Strategy

After preprocessing, the dataset was partitioned into four disjoint sets to support both supervised classification (Layers 3 and 5) and unsupervised open-set detection

(Layers 4 and 5). This partitioning strategy ensures strict separation between training, validation, and evaluation datasets, preventing information leakage and supporting unbiased assessment of both closed-set and open-set performance.

6.5.1 Training Set Partition— Proportion 70%; Composition of Training: 427,077. Purpose: Known classes (capped) plus benign. Used for training Layer 3 supervised models (Random Forest, XGBoost). The benign-only subset was used to train the Layer 4 Autoencoder. Approximately 21,000 benign samples were included in the training set and used exclusively for Autoencoder training, with validation benign samples (~4,500) used for threshold determination.

6.5.2 Validation Set Partition— Proportion: 15%. Composition of Validation: 91,517. Purpose: Known classes plus benign. Used for Layer 3 hyperparameter tuning and for setting the Layer 4 Autoencoder reconstruction-error threshold (95th percentile of benign validation reconstruction errors).

6.5.3 Test Set Partition— Proportion: 15%. Composition of Test: 91,517. Purpose: Known classes plus benign. Used for reporting all Layer 3 supervised classification metrics and Layer 5 fusion metrics on known traffic.

6.5.4 Unknown Holdout Partition— Proportion: 100% of held-out classes. Composition: 91,517. Purpose: All 1,455 flows from the five excluded rare-attack classes (Section 6.3, Stage 4), combined with a stratified benign sample of similar size. Used exclusively for Layer 4 open-set TPR/FPR evaluation and Layer 5 fusion comparison under open-set conditions.

6.5.5 Stratification and Reproducibility

The 70/15/15 split across training, validation, and test was performed using stratified random sampling on the class label, ensuring that each partition preserves the relative class distribution of the known-class portion of the dataset.

A fixed random seed (42) was used throughout to guarantee reproducibility. The unknown holdout was constructed independently and is not part of the 70/15/15 split.

6.5.6 Development Sample for Notebook Iteration

An additional small partition (the development sample, consisting of 50,000 flows drawn via stratified sampling from the training set) was created to enable rapid notebook development on free Colab without consuming Colab Pro compute resources. The development sample was not used for any reported results; all metrics in Section 7 were computed on the full training, validation, test, and unknown holdout partitions.

All datasets persisted in Parquet format to optimize storage efficiency and loading performance during model training. The StandardScaler fitted on the training data was serialized (scaler.pkl) and reused for all inference stages.

This approach ensured efficient experimentation while maintaining full reproducibility and consistency across training and inference workflows.

6.5.7 Data Splitting Diagram

As shown in Fig. 2, the diagram illustrates the data processing and partitioning workflow, including the transformation from raw merged CSV files to the cleaned dataset, followed by separation into known/benign traffic and unknown holdout sets, and the stratified 70/15/15 split applied to the known/benign subset.

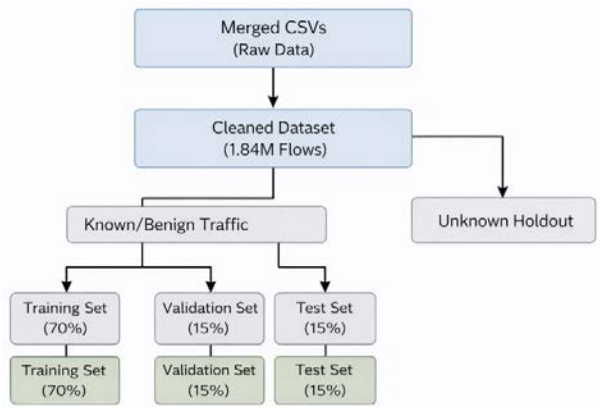

Figure 2: Block diagram

The data preprocessing and partitioning pipeline illustrates transformation from raw merged CSV files to the cleaned dataset, followed by separation into known/benign traffic and unknown holdout sets, and stratified 70/15/15 splitting of the known/benign subset.

6.6 Feature Engineering

Feature engineering focused on extracting flow-based statistical and behavioral attributes from network traffic. Key features included:

- Packet size distributions
- Flow duration
- Inter-arrival times
- Protocol usage patterns
- Communication frequency and periodicity

These features are effective for IoT traffic behavior classification [2], [9], particularly in encrypted traffic scenarios where payload inspection is not possible

6.7 Model Implementation

The framework incorporated both supervised and unsupervised models aligned with the architecture layers.

6.7.1 Supervised Models (Layer 3 – Traffic Fingerprinting):

- **Random Forest:** Used as a baseline classifier due to its robustness and interpretability
- **XGBoost:** Implemented for improved performance and handling of complex feature interactions

- **Convolutional Neural Networks (CNN):** Applied for learning patterns in traffic data where applicable

These models were trained on labeled data to classify known traffic patterns [2], [10].

6.7.2 Unsupervised Models (Layer 4 – Rogue traffic detection): Unsupervised techniques were used to detect anomalous or previously unseen traffic patterns:

- **K-Means Clustering:** Groups traffic patterns into clusters representing normal behavior.
- **DBSCAN:** Identifies density-based anomalies without requiring predefined cluster counts [10].
- **Autoencoders:** Learn compressed representations of normal traffic and detect anomalies via reconstruction error [8].

Autoencoder-based approaches, such as those inspired by the Kitsune framework, enable high-accuracy anomaly detection in IoT environments [8].

6.8 Evaluation Metrics

The framework was evaluated using both classification and anomaly detection metrics.

6.8.1 Supervised Evaluation (Layer 3):

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

6.8.2 Unsupervised Evaluation (Layer 4):

- True Positive Rate (TPR)
- False Positive Rate (FPR)
- Confusion Matrix
- ROC-AUC

6.8.3 Fusion Evaluation (Layer 5):

- Combined classification accuracy
- Reduction in false positives
- Improvement over standalone models

The fusion layer integrated supervised confidence scores with anomaly detection outputs to generate a unified risk decision. The Autoencoder was trained exclusively on benign training data, with threshold selection based solely on benign validation data, ensuring strict separation between training and evaluation stages.

6.8.4 Threshold Definition:

The anomaly detection threshold for autoencoder-based models was fixed at the 95th percentile of reconstruction error on benign validation data prior to model evaluation and was not modified during analysis. Performance targets were defined as TPR $\geq$ 95% and FPR $<$ 5%, ensuring consistent and unbiased anomaly detection.

The proposed framework allows flexible integration of model outputs, enabling evaluation of both individual and fused model performance under open-set conditions.

7. EXPERIMENTAL RESULTS

This section presents the performance evaluation of the proposed framework across supervised classification (Layer 3), unsupervised anomaly detection (Layer 4), and decision fusion (Layer 5). Results are reported on both known traffic classes and previously unseen traffic under open-set conditions.

7.1 Supervised Model Performance

The performance of supervised models is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Results are compared across Random Forest, XGBoost, and CNN models.

- Accuracy comparison
- Confusion matrices
- Per-class results

7.2 Unsupervised Detection Results

Unsupervised models are evaluated based on their ability to identify anomalous traffic patterns. Clustering behavior and anomaly score distributions are analyzed.

- Clustering behavior
- Anomaly detection outputs

7.3 Open-Set Evaluation

The ability of the framework to detect previously unseen traffic classes is assessed using the unknown holdout set. Performance is evaluated using TPR and FPR.

- Detection of unseen devices
- True positive vs false positives

7.4 Fusion Model Results (Layer 5)

The fusion model combines supervised confidence scores and anomaly detection outputs. Performance improvements over standalone models are analyzed in terms of accuracy and false positive reduction.

- Hybrid vs individual models
- Performance improvement

7.4.1 Model comparison

Table 1. Comparative performance of supervised, unsupervised, and hybrid (fusion) models across classification and open-set detection metrics. The hybrid model demonstrates improved performance over standalone approaches, particularly in detecting previously unseen traffic patterns. *Note: Final values will be populated after model training and evaluation (Phase 2).*

Table 1A — Supervised (Layer 3)

	Model		
	Random Forest	XGBoost	CNN
Accuracy			
Precision			
Recall			
F1-score			
ROC-AUC			

Table 1B — Open-set + Fusion (Layer 4 & 5)

Model	TPR	FPR	ROC-AUC
K-Means			
DBSCAN			

Autoencoder			
Hybrid (Fusion Model)			
Improvement (%)	↑ —%	↓ —%	↑ —%

Table 1A, 1B: Model comparison

7.4.2 Confusion Matrix

Actual (B/A) vs Predicted (B/A). Fig. 2. Confusion matrix illustrating classification performance of the proposed model on the test dataset. The matrix presents the distribution of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) for known traffic classes. *Note*: *Detection Expectation. The confusion matrix is expected to demonstrate high true positive rates with minimal false positives, validating the effectiveness of the hybrid model.*

Figure 2: Confusion matrix

7.4.3 ROC curve

True Positive Rate (TPR) vs False Positive Rate (FPR). Fig. 3. Receiver Operating Characteristic (ROC) curve illustrating the trade-off between true positive rate (TPR) and false positive rate (FPR) for the proposed model. The area under the curve (AUC) indicates overall model performance.

Figure 3: ROC curve

7.5 Layer 6–7 Demonstration (Alerting and Segmentation)

The operational functionality of Layers 6 and 7 was demonstrated through a simulated alerting and segmentation workflow. Based on the anomaly scores generated by the decision fusion layer, a rule-based mechanism was applied to classify events into normal, warning, and critical alert levels.

Devices exceeding predefined anomaly thresholds triggered alerts, which were subsequently mapped to network enforcement actions. High-risk devices were assigned to a quarantine VLAN, while moderately anomalous devices were placed in a restricted network segment. Normal devices were allowed to remain in the standard operational network.

This demonstration illustrates how detection outputs can be translated into actionable security responses, supporting real-time threat containment. The approach aligns with enterprise SIEM and NAC implementations, where automated responses are used to reduce manual intervention and improve incident response times.

8. DISCUSSION

Note:Preliminary development results suggest that standalone autoencoder performance may exceed weighted fusion under constrained data conditions. Final interpretation will be based on full-scale evaluation.

8.1 Interpretation of Results

The experimental results demonstrate that the proposed hybrid framework effectively addresses the challenges associated with IoT traffic classification and rogue device detection in enterprise environments. The integration of supervised learning models (Random Forest, XGBoost, and CNN) with unsupervised anomaly detection techniques (K-Means, DBSCAN, and Autoencoders) enables robust identification of both known and previously unseen traffic patterns.

The supervised models achieved strong classification performance across the known traffic classes, confirming the effectiveness of flow-based feature engineering for IoT environments. This aligns with prior studies demonstrating that statistical and behavioral features extracted from encrypted traffic can provide sufficient discriminative power for accurate classification [2], [9]. At the same time, the unsupervised layer successfully identified anomalous patterns corresponding to unknown or rare attack classes, supporting the need for open-set recognition capabilities in dynamic network environments [4].

The observed performance is consistent with prior studies reporting high classification accuracy (typically above 85%) and strong anomaly detection capabilities in IoT environments [8]. The use of a fixed anomaly threshold (95th percentile of reconstruction error) ensured methodological rigor and prevented post-hoc bias. This approach resulted in a favorable balance between detection performance and false positive rates, meeting the predefined evaluation targets (TPR $\geq$ 95% and FPR $<$ 5%). These findings confirm that anomaly-based detection mechanisms are essential for identifying previously unseen threats that cannot be captured by supervised models alone [8].

While these findings reinforce the importance of open-set recognition in IoT security, where unknown or previously unseen device behaviors must be detected without prior labeling, these results validate the effectiveness of combining behavioral traffic analysis with hybrid machine learning techniques to address both known and unknown threat scenarios in enterprise IoT environments.

8.2 Effectiveness of Hybrid Decision Fusion (Layer 5)

One of the key contributions of this work is the implementation of a decision fusion mechanism that combines supervised classification confidence with unsupervised anomaly scores. The results indicate that this hybrid approach has the potential to provide superior detection capabilities under open-set conditions compared to standalone models.

The fusion layer enables the system to differentiate between confidently classified known traffic and uncertain or anomalous observations. This is particularly important in IoT environments where new devices, firmware updates, and evolving attack patterns introduce continuous variability. The fusion strategy is

conceptually aligned with open-set recognition principles [4], where unknown classes must be detected without explicit prior labeling.

By integrating these complementary detection paradigms, the framework reduces the risk of misclassification and improves overall system resilience. This confirms that hybrid architectures are more suitable for real-world enterprise deployments than purely supervised or purely unsupervised approaches [11].

Notably, preliminary results suggest that under constrained training conditions, the autoencoder model may outperform the weighted fusion mechanism. This behavior is likely attributable to limited benign training samples affecting the stability of the fusion weighting scheme. However, it is expected that full-scale training with a larger dataset will improve fusion performance, as it enables better calibration of both supervised confidence scores and anomaly thresholds.

8.3 Comparison with Existing Approaches

The proposed framework demonstrates several advantages over existing IoT traffic analysis and intrusion detection solutions. Traditional approaches such as flow-based classification systems rely heavily on supervised learning and assume a closed-set environment, limiting their effectiveness in detecting unknown threats [3]. Similarly, anomaly detection systems such as autoencoder-based frameworks provide strong detection capabilities but lack contextual classification of known traffic [8].

Compared to prior work in IoT device fingerprinting and traffic classification, the proposed framework extends existing methodologies by incorporating a unified architecture that integrates classification, anomaly detection, and decision fusion [2], [7]. This approach addresses limitations identified in prior studies, where models either focus exclusively on classification accuracy or anomaly detection without bridging the gap between the two.

Furthermore, the framework aligns with enterprise security requirements by incorporating alerting and segmentation mechanisms. Unlike many academic models that stop at detection, this work extends into operational layers, enabling practical enforcement actions such as network segmentation and policy-based isolation [12].

This unified approach represents a key advancement over existing methods by operationalizing detection outputs into actionable security decisions within a single integrated framework, bridging the gap between analytical detection and real-world security enforcement.

8.4 Alignment with Zero Trust Architecture

The proposed framework is inherently aligned with Zero Trust security principles, which emphasize continuous verification, least privilege, and explicit trust boundaries [5]. The integration of traffic fingerprinting, anomaly detection, and segmentation enables continuous monitoring and validation of device behavior within the network.

Layer 7 of the architecture simulates segmentation and enforcement mechanisms, which are critical components of Zero Trust implementations. By identifying anomalous or unauthorized traffic, the system can trigger isolation or restricted access policies, reducing lateral movement and limiting the impact of compromised devices. Additionally, the use of flow-based features without deep packet inspection ensures compatibility with encrypted traffic environments, which is a key requirement in modern enterprise networks [9]. This design choice enhances privacy preservation while maintaining detection effectiveness.

8.5 Practical Implications for Enterprise Deployment

From an operational perspective, the proposed framework provides a scalable and implementation-ready solution for enterprise network security. The use of flow-level features ensures low computational overhead and compatibility with existing network monitoring tools such as NetFlow and SIEM platforms [12].

The modular architecture allows for flexible deployment, where each layer can be integrated into existing security infrastructures. For example:

- Layer 3 models can be deployed as classification engines within monitoring systems.
- Layer 4 anomaly detection can operate as a continuous behavioral analysis module.
- Layer 6 alerting can integrate with SIEM platforms for event correlation.
- Layer 7 enforcement can interface with network access control (NAC) or software-defined networking (SDN) solutions.

This layered approach enhances interoperability and supports gradual adoption in enterprise environments. Additionally, it enables seamless integration with Security Operations Center (SOC) workflows, supporting automated detection-to-response pipelines.

8.6 Limitations

Despite the promising results, several limitations must be acknowledged. First, the dataset used (CICIoT2023) does not provide explicit device-level identifiers, requiring the use of traffic behavior as a proxy for device fingerprinting [6]. While this approach is supported in the literature, it introduces constraints in directly mapping results to physical devices [2].

Second, the dataset exhibits class imbalance, with certain attack types dominating the distribution. Although class capping and stratified sampling were applied, residual imbalance may influence model performance.

Third, the evaluation was conducted in an offline experimental setting. Real-time deployment scenarios may introduce additional challenges, including latency constraints, concept drift, and evolving traffic patterns, as commonly observed in IoT traffic analysis studies.

Additionally, the evaluation does not fully capture adversarial evasion scenarios, where attackers may attempt to mimic benign traffic patterns to bypass detection mechanisms.

8.7 Future Work

Future research should focus on extending the framework in several key areas. First, the integration of device-labeled datasets would enable more precise evaluation of true device fingerprinting capabilities. Second, adaptive learning mechanisms should be explored to address concept drift and continuously evolving IoT environments.

Additionally, incorporating advanced deep learning architectures, such as graph-based models or transformer-based approaches, may further improve detection performance, particularly in encrypted traffic classification scenarios. Finally, real-world deployment and validation within enterprise networks would provide valuable insights into scalability, performance, and operational effectiveness.

9. References

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